

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1513
Additional Practice Set 3

1.

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ -1 & 3 & 6 \\ 2 & 1 & 0 \\ 3 & 1 & -1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 1 & 4 & 1 & 2 \\ 4 & 2 & 2 & 3 & 2 \\ 2 & 1 & -2 & 2 & 0 \\ 6 & 3 & 6 & 4 & 4 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 1 & 4 & 5 & 8 \\ -1 & 4 & 3 & 0 \\ 2 & 0 & 2 & 1 \end{pmatrix}.$$

For each matrix, find

- (i) a basis for the row space and a basis for the column space;
- (ii) a basis for the nullspace;
- (iii) the rank and nullity of the matrix, and verify the Dimension Theorem for Matrices.

2. A 3×3 matrix $\mathbf{A}$ has a row echelon form $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$. Determine each of the following if possible. Give brief explanations how you derive the answers.

- (i) a basis for the row space of $\mathbf{A}$;
- (ii) a basis for the column space of $\mathbf{A}$;
- (iii) a basis for the column space of $\mathbf{A}^T$;
- (iv) a basis for the nullspace of $\mathbf{A}$;
- (v) the rank and nullity of $\mathbf{A}$.

3. In each of the following parts, $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are vectors in $\mathbb{R}^3$ that have their initial points at the origin. Determine in each case whether the three vectors lie in a plane.

- (a) $\mathbf{v}_1 = (2, -2, 0), \mathbf{v}_2 = (6, 1, 4), \mathbf{v}_3 = (2, 0, -4)$.
- (b) $\mathbf{v}_1 = (-6, 7, 2), \mathbf{v}_2 = (3, 2, 4), \mathbf{v}_3 = (4, -1, 2)$.

4. Express each of the following set of vectors in $\mathbb{R}^4$ as a linear span and determine the dimension of and a basis for each of the subspaces.

- (a) The set of all vectors in $\mathbb{R}^4$ of the form $(a, b, a - b, a + b)$ for any $a, b \in \mathbb{R}$.
- (b) The set of all vectors in $\mathbb{R}^4$ where all the four components are the same.

5. (i) The two linear equations $x + y = 0$ and $y + z = 0$ are represented by two planes in the xyz -space. These two planes intersect at a line. Express this line in terms of a linear span in $\mathbb{R}^3$.

- (ii) Find a 3×3 matrix $\mathbf{B}$ such that the nullspace of $\mathbf{B}$ is given by the linear span in part (i). (Hint: Use the two equations in part (i).)
6. Consider the three vectors $\mathbf{u} = (1, -1, 0, 0)$, $\mathbf{v} = (0, 1, 1, 0)$, $\mathbf{w} = (0, 0, 0, 1)$ in $\mathbb{R}^4$.
- (i) Are $\mathbf{u}, \mathbf{v}, \mathbf{w}$ linearly independent? Show your working.
- (ii) Find a vector that is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$ but does not belong to $\text{span}\{\mathbf{w}\}$. Justify your answer.
7. An engineer has found two solutions to a homogeneous system $\mathbf{Ax} = \mathbf{0}$ of 40 equations and 42 variables. These two solutions are not scalar multiple of each other, and every solution of the system can be constructed by a linear combination of these two solutions.
- Suppose now the engineer needs to replace the homogeneous system with an associated non-homogeneous system $\mathbf{Ax} = \mathbf{b}$ for some non-zero vector $\mathbf{b}$. Can the engineer be certain that he will be able to find a solution?
8. The regular coordinate axes for the xyz -space is being transformed as follow: a rotation about the z -axis through 60° counterclockwise, viewing from the positive z -axis, followed by a reflection of the z -axis about the xy -plane.
- (i) Find a basis S consisting of unit vectors that determine the new coordinate axes.
- (ii) What are the coordinates of the vector $\mathbf{v} = \begin{pmatrix} 4 \\ 0 \\ 3 \end{pmatrix}$ relative to the new axes?
- (iii) Find a matrix $\mathbf{M}$ such that $\mathbf{v} = \mathbf{M}(\mathbf{v})_S$, where $(\mathbf{v})_S$ is the coordinate vector of $\mathbf{v}$ written in column form relative to the basis S .
9. The position y of a moving object in a straight line was measured in meters every second from time $t = 0$ to 4 second, given by 0, 1, 3, 6 and 10 metre with respect to the starting point.
- (i) Find the least squares quadratic curve $y = a + bt + t^2$ for these data.
- (ii) Estimate the position and velocity of the object when $t = 5$.
10. Two vectors $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \in \mathbb{R}^3$ lie on a plane P through the origin in the xyz -space.
- (i) Write down the linear span form that represents the plane P .
- (ii) Write down a linear equation that represents P .
- (iii) Let $\mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$. Show that $\mathbf{b}$ does not lie on P .

- (iv) Find the least squares solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ where $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 1 & 1 \\ 0 & 2 \end{pmatrix}$ and $\mathbf{b}$ is as in (iii).
- (v) Find the projection $\mathbf{w}$ of $\mathbf{b}$ on the plane P . (Hint: use (iv).)
- (vi) Find the coordinate vector of the projection $\mathbf{w}$ in (v) with respect to the basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ for the plane P .
- (vii) Find the shortest distance from the point $(1, 0, 0)$ to the plane P .